Implementing a central twist on two permutation domains

Human Compression project: F006 continuation

The complete proof of *Enumeration by stabilizers of initial strings* remains a prerequisite. Its orbit recursion, intersection test and necessary completion test are unchanged. This note replaces a large permutation representation of the automorphism group by a smaller conjugation action. It makes no claim that the exceptional census is complete.

Lemma 1. *Let $G \leq \mathrm{Sym}(\Omega)$ be finite. Suppose $z \in Z(G)$ has order two and $\chi : G \to \{0,1\}$ is an additive character with $\chi(z) = 0$. Then $\phi(g) = z^{\chi(g)}g$ is an involutory automorphism. On $\Omega \times \{0,1\}$ define*

$$\iota(g)(x,0) = (g(x),0), \qquad \iota(g)(x,1) = (\phi(g)(x),1), \qquad s(x,i) = (x,1-i).$$

The map ι is an injective homomorphism, and conjugation by s implements ϕ on $\iota(G)$. Put $A = \langle \iota(G), s\rangle$. The image of the conjugation action of A on $\iota(G)$ is exactly $\langle \mathrm{Inn}(G), \phi\rangle$.

Proof. Centrality and additivity give $\phi(gh) = \phi(g)\phi(h)$. Since $\chi(z) = 0$, applying ϕ twice returns g. Thus it is an automorphism, including when it is the identity. The two coordinate actions define a homomorphism; its first coordinate is the given faithful action of G, so the homomorphism is injective. Swapping the two domains interchanges the actions g and $\phi(g)$. Because $\phi^2 = 1$, this gives $s^{-1}\iota(g)s = \iota(\phi(g))$. Consequently $\iota(G)$ is normal in A. Conjugation by its elements gives the inner automorphisms, and conjugation by the additional generator gives ϕ; these generate exactly the stated image. □

Proposition 2. *Suppose exact group operations establish*

$$|A : C_A(\iota(G))| = |\,\mathrm{Aut}(G)|. \tag{1}$$

In the retained enumeration algorithm, one may use A acting by conjugation on the involutions of $\iota(G)$. For an already chosen tuple t, use $C_A(t)$ in place of its automorphism stabilizer. The resulting search records precisely the same abstract generating string C-markings, up to isomorphism, as the full automorphism action. The same statement applies to its reversal and connected-string counts.

Proof. Normality makes the conjugation image a subgroup of the full automorphism group. Its kernel is exactly $C_A(\iota(G))$, so (1) makes the image the whole finite automorphism group. The map $A \to \mathrm{Aut}(G)$ is therefore surjective. For each tuple t, the inverse image of its pointwise automorphism stabilizer is exactly $C_A(t)$: membership means fixing every chosen group element under conjugation. Hence both groups have the same orbits on all candidate involutions. This remains true at every successive prefix. The retained completeness proof applies without change. The nontrivial kernel acts trivially throughout; it neither creates nor removes a candidate orbit. Surjectivity also gives the same existence test for an automorphism carrying a full tuple to its reversal. Connectedness depends only on the orders of adjacent products and is preserved by ι. □

The actual finite guards. The implementation takes χ to be permutation parity. It checks that the original model has its stated order, that its centre has order two, and that the unique nonidentity central element is even. It builds the explicit permutations above, verifies their generated order, normality, and conjugation identity on every original generator, and checks (1) against the exact order returned by GAP's full automorphism computation. It then uses the unchanged conjugation recursion and completion pruning. A failure of any guard stops the search; no assertion that this twist always supplies every outer automorphism is used.

For the retained E_8 reflection model the guarded calculation gives original degree 240, new degree 480, group order 696729600, overgroup order 1393459200, conjugation kernel order 2 and full automorphism order 696729600. This avoids constructing its automorphism permutations on all 199951 involutions. Those involutions still form the complete candidate set. The original model constructor, exact group operations and complete census output remain required computational premises. The group orders in this paragraph are checked finite facts, not replacements for the guards or for the still unfinished census.

As a small control, take $G = \langle (1\ 2), (1\ 2\ 3), (4\ 5)(6\ 7) \rangle$. The same construction gives degree fourteen and a full conjugation action of order twelve. Both algorithms enumerate its entire seven-involution set and the same complete generating markings: one ordered class of rank two and two of rank three. This is a check on the implementation; the uniform justification is the two proofs above.

Attribution and limits. The central automorphism and induced permutation action are standard group constructions; no priority is claimed for them. Their explicit guarded use here is an implementation improvement for the retained stabilizer search, whose Hartley–Hulpke and Leemans–Mixer provenance remains unchanged. A complete exceptional-group computation and distinct mathematical review are still required. This note supplies no whole-original, open-question, Lean or compression-factor credit.